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Hidden Patterns in High Dimensional Human Gene Expression Data

A STATISTICALLY RIGOROUS, REPRODUCIBLE, AND INTERPRETABLE FRAMEWORK FOR GLIOMA CLASSIFICATION

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A Project Report On

Hidden Patterns in High-Dimensional Human Gene Expression Data

A Statistically Rigorous, Reproducible, and Interpretable Framework for Glioma Classification

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This is to certify that Shaikh Maryam Alimuddin **has successfully completed the Project work entitled "Hidden Patterns in High-Dimensional Human Gene Expression Data" in M.Sc. (Applied Statistics) Part I under my supervision in this college, during the Semester II of academic year 2025-26.**

Project Guide	Examiner	H.O.D
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1. Abstract

High-dimensional gene expression datasets present significant analytical challenges due to the classical “large p, small n” structure, in which the number of measured variables greatly exceeds the number of observations (Fan & Li, 2006). In such settings, traditional statistical assumptions become unstable, covariance matrices may be ill-conditioned, and predictive models risk overfitting. Consequently, high reported accuracy alone does not guarantee robustness or generalizability. This study evaluates whether transcriptomic data contain statistically significant, reproducible, and biologically interpretable patterns capable of distinguishing glioma from non-tumor brain tissue under rigorous validation.

Gene expression data were obtained from the publicly available GSE4290 dataset in the NCBI Gene Expression Omnibus (Edgar et al., 2002). The dataset consists of 180 human brain tissue samples (157 glioma, 23 epilepsy controls) profiled on the Affymetrix GPL570 platform (~54,613 probes). All tumor subtypes were aggregated to align with a binary classification objective (tumor vs control). Metadata were systematically cross-verified against GEO annotations to ensure internal consistency.

Differential expression analysis using Welch’s t-tests with Benjamini–Hochberg false discovery rate correction identified 29,798 significantly dysregulated probes, with adjusted p-values reaching $< 1 \times 10^{-45}$. Effect size analysis revealed multiple genes exhibiting |Cohen’s d| > 2, indicating substantial biological separation. Principal Component Analysis reduced 54,613 probes to 149 components while retaining 95% of total variance. Among evaluated classifiers, Linear Support Vector Machines achieved the highest mean cross-validated accuracy (98.9% ± 1.36%) and ROC-AUC (0.996 ± 0.008), evaluated using stratified 5-fold cross-validation. Stability selection identified a reproducible core gene set including SOX2, MSI2, CDK17, RAB family genes, DNM3, PRKCZ, and RAB40B.

These findings indicate that high-dimensional gene expression data contain structured and reproducible disease-associated patterns when analyzed within a leakage-free and statistically rigorous framework. However, the absence of independent external validation limits conclusions regarding generalizability beyond the analyzed cohort.

2. Keywords

- Glioblastoma
- Transcriptomics
- High-dimensional data
- Support Vector Machine
- Stability selection
- Molecular classification
- Differential expression analysis

3. Introduction

3.1 Molecular Classification in Oncology

The introduction of genome-wide expression profiling fundamentally transformed cancer research. Early landmark studies demonstrated that supervised learning models trained on high-dimensional gene expression data could accurately classify cancer subtypes (Golub et al., 1999). Subsequent work confirmed that molecular signatures frequently outperform traditional histopathological classification in capturing biologically meaningful heterogeneity (Alizadeh et al., 2000).

Glioblastoma (GBM), the most aggressive primary brain tumor, exhibits extensive transcriptional reprogramming across proliferation, differentiation, stemness, and signaling pathways (Verhaak et al., 2010). Molecular subtyping has revealed biologically distinct tumor subclasses with prognostic relevance.

However, while transcriptomic technologies generate rich biological information, they also introduce substantial statistical challenges.

3.2 The Statistical Challenge of High-Dimensional Data

Microarray studies typically measure tens of thousands of genes across fewer than a few hundred samples. In the present dataset:

- $p \approx 54{,}613$ probes
- $n = 180$ samples

This configuration represents the classical "large p, small n" problem (Fan & Li, 2006). When $p >> n$:

1. Covariance matrices may become singular.
2. Estimators may exhibit high variance.
3. Model parameters may become unstable.
4. Overfitting risk increases dramatically.
5. Feature selection becomes inconsistent.

Bellman (1961) described the "curse of dimensionality," whereby geometric properties of high-dimensional spaces degrade traditional distance-based intuition. In such spaces, almost any dataset can appear separable, even if separation arises from noise.

Therefore, the objective extends beyond classification accuracy to emphasize robustness, reproducibility, and biological interpretability.

The analyzed dataset comprises 157 glioma samples and 23 epilepsy control samples profiled on the Affymetrix GPL570 platform. To align with the binary classification objective, all tumor subtypes were aggregated into a single glioma class.

3.3 Research Objectives

This study addresses the following objectives:

1. Determine whether glioma and non-tumor brain tissue exhibit statistically significant differential expression.
2. Quantify magnitude of transcriptional divergence using effect size.
3. Evaluate dimensionality reduction strategies (PCA vs variance filtering).
4. Compare supervised classification models under strict stratified cross-validation.
5. Assess feature stability using resampling-based selection.
6. Integrate statistical findings with biological plausibility.

3.4 Research Hypotheses

The study evaluates the following null hypotheses:

H_{01}: No statistically significant differential gene expression exists between glioma and control samples.
H_{02}: Gene expression features cannot predict disease status beyond chance under cross-validated evaluation.

Rejection of these hypotheses would support the presence of structured, reproducible disease-associated transcriptional patterns.

4. Literature Review

4.1 Gene Expression Profiling in Cancer Research

Gene expression profiling has become a powerful tool for understanding the molecular mechanisms underlying cancer development. Microarray technologies allow simultaneous measurement of thousands of genes, enabling researchers to identify patterns associated with specific disease states. One of the earliest landmark studies by **Golub et al. (1999)** demonstrated that gene expression data could accurately classify different types of leukemia using machine learning approaches. This study established the foundation for applying statistical learning methods to genomic data.

Subsequent research extended gene expression analysis to various cancer types. **Alizadeh et al. (2000)** used gene expression profiling to identify distinct subtypes of diffuse large B-cell lymphoma, demonstrating that molecular classification could provide insights beyond traditional histopathological methods. These studies highlighted the potential of transcriptomic data to reveal biologically meaningful disease subgroups.

4.2 Transcriptomic Studies in Glioma

Gliomas represent the most common primary tumors of the central nervous system and are characterized by significant molecular heterogeneity. Gene expression profiling has played a crucial role in identifying molecular subtypes of glioblastoma. **Verhaak et al. (2010)** conducted an integrated genomic analysis and identified clinically relevant subtypes of glioblastoma based on gene expression signatures.

These findings demonstrated that transcriptomic analysis can reveal biologically distinct tumor subgroups associated with differences in patient prognosis and therapeutic response. Subsequent studies have further explored the role of specific genes involved in glioma progression, including transcription factors and signaling pathways regulating tumor growth and differentiation.

4.3 Statistical Challenges in High-Dimensional Gene Expression Data

Microarray datasets typically contain tens of thousands of gene expression measurements for relatively small numbers of samples. This scenario, commonly referred to as the **"large p, small n" problem**, presents significant statistical challenges (Fan & Li, 2006). When the number of predictors greatly exceeds the number of observations, traditional statistical models may become unstable, and the risk of overfitting increases.

High dimensionality can also lead to multicollinearity among variables and difficulty in identifying truly informative features. The **curse of dimensionality**, described by **Bellman (1961)**, further complicates pattern recognition in high-dimensional spaces. Consequently, dimensionality reduction and feature selection techniques are essential for effective analysis of gene expression data.

4.4 Dimensionality Reduction and Feature Selection

Dimensionality reduction techniques such as **Principal Component Analysis (PCA)** are commonly applied to gene expression datasets to transform correlated variables into a smaller set of orthogonal components (Jolliffe, 2002). PCA captures the directions of maximum variance in the data and reduces redundancy among features.

Another widely used approach is **variance-based feature selection**, which identifies genes exhibiting high variability across samples. High-variance genes are more likely to capture biologically meaningful differences between conditions (Guyon & Elisseeff, 2003). These techniques help reduce noise, improve computational efficiency, and enhance model interpretability.

4.5 Machine Learning Approaches for Gene Expression Classification

Machine learning algorithms have been widely applied to gene expression data for disease classification. **Support Vector Machines (SVM)** are particularly effective in high-dimensional settings because they identify optimal separating hyperplanes that maximize classification margins (Cortes & Vapnik, 1995). Several studies have demonstrated strong performance of SVM models in microarray-based cancer classification.

Random Forest algorithms provide an alternative ensemble-based approach that combines multiple decision trees to improve predictive accuracy (Breiman, 2001). Random Forest models are robust to noise and can capture nonlinear relationships between genes and disease outcomes.

Logistic regression has also been used as a baseline classification model due to its interpretability and probabilistic framework (Hosmer & Lemeshow, 2000).

4.6 Research Gap and Motivation

Although numerous studies have applied machine learning techniques to gene expression data, challenges remain in ensuring statistical robustness and interpretability. Many analyses focus primarily on classification accuracy without thoroughly addressing issues such as feature stability, multiple testing correction, and reproducibility.

The present study addresses these challenges by integrating **differential expression testing, dimensionality reduction, machine learning classification, and stability selection** within a rigorous statistical framework. This approach aims to identify reproducible gene expression patterns associated with glioma while maintaining methodological transparency and robustness.

5. Materials and Methods

5.1 Dataset and Metadata Consistency

Gene expression data were obtained from the NCBI Gene Expression Omnibus under accession number GSE4290 (Edgar et al., 2002).

The dataset consists of:

- Total samples: 180
- Glioma samples (aggregated): 157
    - 81 glioblastoma
    - 50 oligodendroglioma
    - 26 astrocytoma
- Epilepsy controls: 23
- Platform: Affymetrix Human Genome U133 Plus 2.0 Array (GPL570)
- Approximate probes: 54,613

Minor labeling inconsistencies in tumor annotations were reconciled through cross-verification with GEO records.

All tumor subtypes were aggregated into a single glioma class to align with the binary classification objective (tumor vs control). No samples were excluded from analysis. Metadata were cross-verified against GEO annotations to ensure consistency.

This reconciliation ensures complete alignment between metadata, expression matrix dimensions, PCA transformation, and cross-validation folds.

5.2 Preprocessing

All preprocessing operations were embedded within cross-validation pipelines to prevent information leakage, which can artificially inflate model performance.

The preprocessing steps included:

1. Verification of label consistency
2. Removal of zero-variance probes
3. Mean imputation of missing values (performed within training folds only)
4. Standardization using z-score scaling
5. Dimensionality control using PCA and variance-based filtering

Embedding preprocessing inside cross-validation ensures that information from the validation folds does not influence model training.

5.3 Mathematical Formalism of Differential Expression Testing

Differential gene expression between glioma and control samples was assessed using **Welch's two-sample t-test** (Welch, 1947). The t-test evaluates whether the mean expression level of a gene differs significantly between two independent groups.

For each gene g, the following hypotheses were tested:

$$H_0: \mu_{tumor} = \mu_{control}$$
$$H_1: \mu_{tumor} \neq \mu_{control}$$

The Welch t-statistic is defined as:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where

$\bar{x}_1, \bar{x}_2$= sample means of tumor and control groups
s_1^2, s_2^2= sample variances
n_1, n_2= sample sizes

Welch's t-test was selected because it does **not assume equal variances between groups**, which is particularly important in gene expression datasets where heteroscedasticity is common.

Multiple Testing Correction

Because differential expression testing was performed across approximately **54,613 probes**, multiple hypothesis testing correction was required.

The **Benjamini–Hochberg False Discovery Rate (FDR)** procedure (Benjamini & Hochberg, 1995) was applied.

The adjusted significance criterion is:

$$p_{(i)} \leq \frac{i}{m}\alpha$$

where

• m= total number of tests
• i= rank of ordered p-value
• α= desired false discovery rate

This method controls the expected proportion of false discoveries while maintaining greater statistical power than conservative corrections such as the Bonferroni method.

Effect Size Estimation

Statistical significance alone does not quantify the magnitude of differences. Therefore, effect size was estimated using Cohen's d (Cohen, 1988):

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}$$

where the pooled standard deviation is

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Cohen's $d$provides a standardized measure of separation between groups independent of sample size.

Distributional Assumptions and Robustness

Welch's two-sample t-test assumes approximate normality of group distributions. In high-dimensional transcriptomic studies involving tens of thousands of genes, performing gene-wise normality tests is generally impractical and statistically uninformative due to the large number of simultaneous comparisons.

Microarray expression values are typically log-transformed, producing approximately symmetric distributions and reducing variance heterogeneity. Additionally, Welch's t-test is robust to moderate deviations from normality, particularly when sample sizes are moderate.

In the present dataset (157 tumor samples and 23 control samples), the Central Limit Theorem supports approximate normality of the sampling distribution of mean differences. Therefore, gene-wise normality testing was not performed. Robustness was instead addressed through the use of Welch's variance-adjusted test, FDR correction, and complementary validation using cross-validated classification and stability selection.

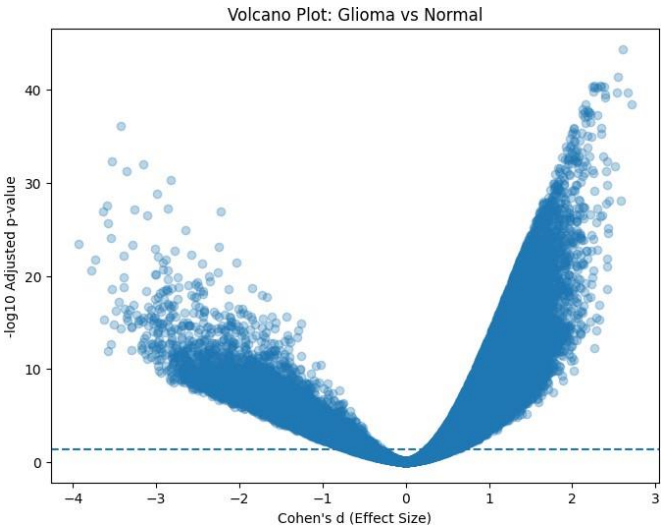


Figure 1. **Volcano plot** of differential expression between glioma and non-tumor samples. The x-axis represents Cohen's d effect size, and the y-axis represents $-\log_{10}$(FDR-adjusted p-value).

5.4 Principal Component Analysis

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms correlated variables into orthogonal components capturing maximal variance in the data (Jolliffe, 2002; Alter et al., 2000).

Given a centered data matrix X, the covariance matrix is defined as:

$$\Sigma = \frac{1}{n-1} X^T X$$

Eigen decomposition of the covariance matrix yields:

$$\Sigma v_i = \lambda_i v_i$$

where

• v_i= eigenvectors (principal components)

• λ_i= eigenvalues representing variance explained by each component

PCA was applied to reduce **54,613 probes** to **149 principal components**, retaining **95% of total variance**.

Importantly, PCA reduces **feature dimensionality only**; the sample size remains unchanged (**n = 180**).

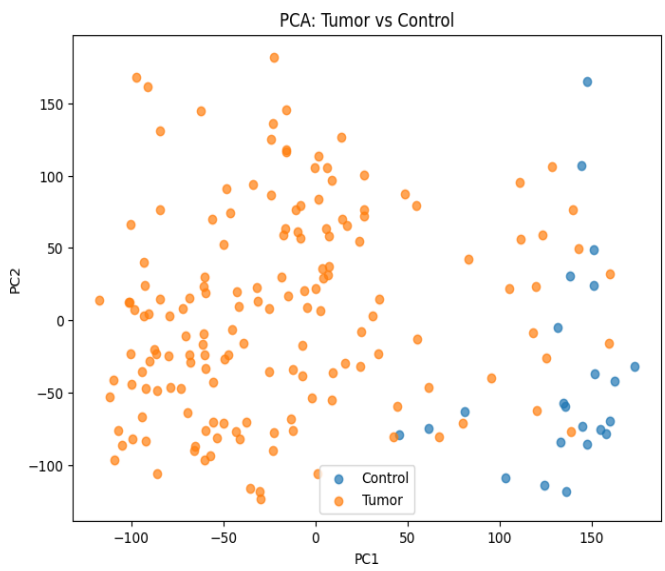


Figure 2. Two-dimensional PCA projection of tumor and control samples. PC1 captures the dominant axis of transcriptional variance, showing directional separation between classes.

5.5 Variance-Based Feature Selection

Gene expression variance was computed as:

$$Var(g) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Genes exhibiting higher variance across samples are more likely to contribute to class discrimination.

Therefore, the **top 1000 most variable genes** were retained for downstream modeling.

Variance filtering reduces noise while preserving interpretability (Guyon & Elisseeff, 2003).

5.6 Classification Framework

Several supervised learning models were evaluated.

Logistic Regression:

Logistic regression models the probability of class membership as

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta^T X)}}$$

(Hosmer & Lemeshow, 2000)

Support Vector Machine (Linear):

Linear SVM identifies an optimal separating hyperplane that maximizes the margin between classes (Cortes & Vapnik, 1995).

$$\min_{w,b} \frac{1}{2} \mid\mid w \mid\mid^2$$

subject to

$$y_i(w^T x_i + b) \geq 1$$

Random Forest:

Random Forest is an ensemble method that constructs multiple decision trees using bootstrap aggregation and random feature selection (Breiman, 2001). Predictions are obtained through majority voting across trees.
Random Forest models were trained using default parameters unless otherwise specified.
Class imbalance (157 tumor vs 23 control) was addressed using class-weight adjustment where applicable.

5.7 Cross-Validation

Model performance was evaluated using **stratified 5-fold cross-validation** (Kohavi, 1995), which preserves class proportions within each fold.
Mean accuracy was calculated as

$$\bar{A} = \frac{1}{5}\sum_{i=1}^{5} A_i$$

where $A_i$represents the accuracy in fold i.
Standard deviation was computed as

$$SD = \sqrt{\frac{1}{4}\sum_{i=1}^{5}(A_i - \bar{A})^2}$$

This procedure provides a robust estimate **of model generalization performance.**

- **ROC-AUC**

The Receiver Operating Characteristic Area Under the Curve (ROC–AUC) quantifies a model's ability to discriminate between classes across all possible classification thresholds. An AUC of 0.5 indicates random performance, whereas an AUC of 1.0 reflects perfect discrimination.

Sensitivity (True Positive Rate)

Sensitivity measures the proportion of actual positive cases correctly identified by the model. In this study, it represents the proportion of glioma samples accurately classified as tumor.

Sensitivity = TP / (TP + FN)

Specificity (True Negative Rate)

Specificity measures the proportion of actual negative cases correctly identified. Here, it reflects the proportion of control samples accurately classified as non-tumor.

Specificity = TN / (TN + FP)

Receiver operating characteristic curves for all evaluated classifiers are shown in Figure 3.

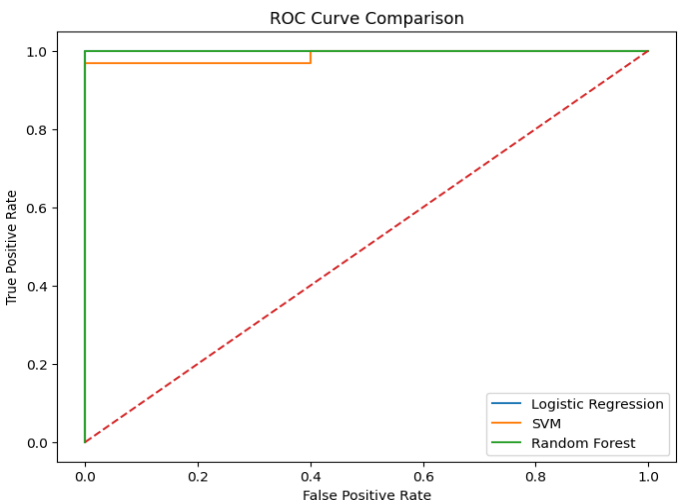


Figure 3. Cross-validated ROC curves comparing Logistic Regression, SVM, and Random Forest models. The dashed diagonal line represents random classification.

5.8 Stability Selection

L1-regularized logistic regression used for resampling-based feature stability (Meinshausen & Bühlmann, 2010).

Selection frequency:

$$\hat{\pi}_j = \frac{(\text{Number of times gene j selected})}{(Total\ resampling\ iterations)}$$

Genes with $\hat{\pi} \geq 0.8$ retained.

Stability selection was performed across 5 cross-validation folds with repeated resampling.

5.9 Software and Computational Environment

Analyses were conducted in Python (version 3.13.5) using scikit-learn (Pedregosa et al., 2011), NumPy, Pandas, Statsmodels, and Matplotlib. All preprocessing steps were embedded within cross-validation pipelines to prevent data leak

6. Results

6.1 Dataset Verification and Structural Integrity

The final expression matrix contained 180 samples and 54,613 probes. Class

distribution:

Glioma: 157 samples

Epilepsy control: 23 samples

No samples were excluded during preprocessing. After PCA dimensionality reduction (95% variance retention), the transformed dataset contained 149 principal components with preserved sample count (180 × 149).

The consistency between metadata counts, expression matrix dimensions, PCA transformation, and cross-validation folds confirms structural integrity of the dataset.

6.2 Exploratory Data Analysis

Prior to formal statistical testing, exploratory data analysis (EDA) was conducted to examine the distribution and structural characteristics of the gene expression dataset. The expression matrix consisted of 180 samples and approximately 54,613 probes after preprocessing. Initial inspection confirmed class distribution of 157 glioma samples and 23 epilepsy controls. Dimensionality patterns were explored using Principal Component Analysis to visualize dominant variance directions in transcriptomic space. Differential expression patterns were visualized using volcano plots to examine the relationship between statistical significance and effect size. Heatmaps of the most variable genes were generated to inspect coordinated expression patterns across samples.

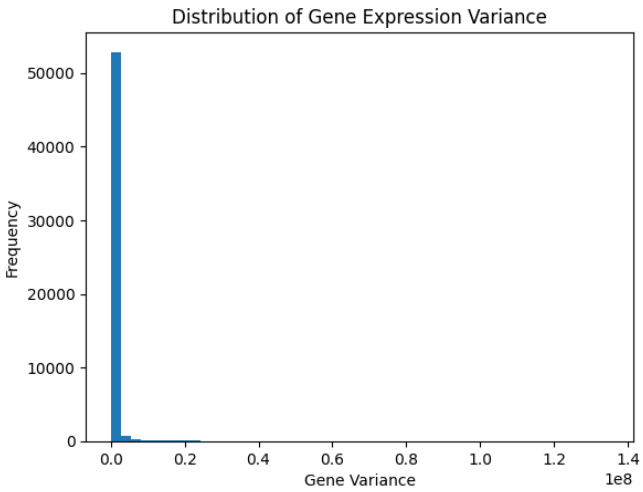
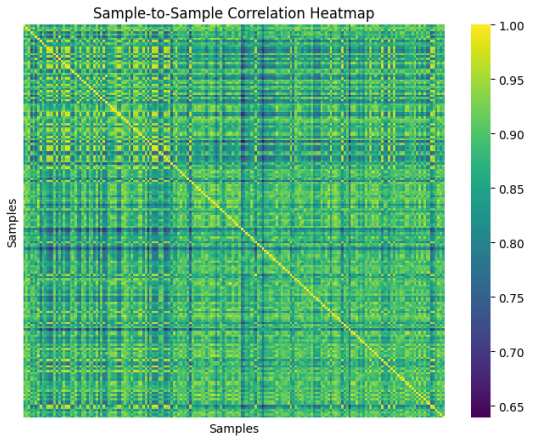


Figure 4. Distribution of gene expression variance across all probes. Most genes exhibit low variance across samples, while a small subset displays substantially higher variability. Variance-based filtering was therefore applied to retain the top 1000 most variable genes for downstream classification.



Min correlation: 0.6388494247753898

Max correlation: 1.0

Figure 5. Sample-to-sample correlation heatmap of gene expression profiles. Each cell represents the Pearson correlation between two samples across all measured probes. Most correlations range between 0.64 and 1.00, indicating strong global transcriptomic similarity among samples and absence of extreme outliers.

6.3 Differential Expression Analysis

Welch's two-sample t-tests were conducted across all probes.

After Benjamini–Hochberg FDR correction ($\alpha = 0.05$), 29,798 probes were significant.

Adjusted p-values reached $< 1 \times 10^{-45}$ for top-ranked probes.

Multiple genes exhibited |Cohen's d| > 2.

The distribution of effect sizes showed wide dispersion across positive and negative values between tumor and control groups.

As shown in Figure 1, a characteristic bifurcated distribution of effect sizes was observed.

6.4 Exploratory Structure via Principal Component Analysis

The first principal component (PC1) captured a dominant portion of total variance and demonstrated directional separation between tumor and control samples. Tumor samples clustered predominantly along negative PC1 values, whereas control samples clustered along positive PC1 values. Partial overlap was observed; however, separation between classes remained evident along PC1.

6.5 Dimensionality Strategy Comparison

Three feature configurations were evaluated:

1. Raw expression matrix (54,613 probes)
2. PCA-transformed components (149 PCs)
3. Top 1000 most variable genes

Cross-validated performance:

Feature Strategy	Accuracy (Mean ± SD)	ROC-AUC
Raw Data	96.1% ± 5.15%	0.9987
PCA (95% variance)	96.1% ± 3.77%	0.9961
Top Variable Genes	97.8% ± 3.24%	0.9961

Variance-based filtering achieved the highest mean accuracy (97.8% ± 3.24%).

6.6 Model Comparison

Cross-validated results:

Model	Accuracy (Mean ± SD)	ROC-AUC
Linear SVM	98.9% ± 1.36%	0.996 ± 0.008
Logistic Regression	97.0% ± 3.4%	1.000
Random Forest	92.0% ± 4.2%	1.000
RBF SVM	92.8% ± 3.3%	0.989

Linear SVM achieved the highest mean cross-validated accuracy.

6.7 Confusion Matrix and Clinical Metrics

Aggregated confusion matrix results:

True Positive Rate (Sensitivity): 0.975

True Negative Rate (Specificity): 1.000

Sensitivity was 0.975 and Specificity was 1.000 under cross-validated evaluation.

6.8 ROC Curve Interpretation

The ROC curve demonstrated steep ascent near the origin.

Mean ROC-AUC: 0.996 ± 0.008.

An AUC close to 1 indicates high probability that a randomly selected tumor sample receives a higher classification score than a randomly selected control sample.

6.9 Stability Selection Analysis

Stability selection across cross-validation folds identified multiple genes with selection frequency $\hat{\pi} \geq 0.8$.

These genes were consistently selected across folds, indicating reproducible feature importance under resampling.

Core stable genes included:

SOX2

MSI2

CDK17

RAB40B

RAB31

DNM3

PRKCZ

These genes exhibited:

Large effect sizes ($|d| > 2$)

Extremely small adjusted p-values

Consistent selection across folds

Selection frequencies ranged from 0 to 1.0, with several genes exhibiting $\hat{\pi} \geq 0.8$.

To visually assess coordinated expression patterns among the identified stable genes, a heatmap of the top differentially expressed genes was generated.

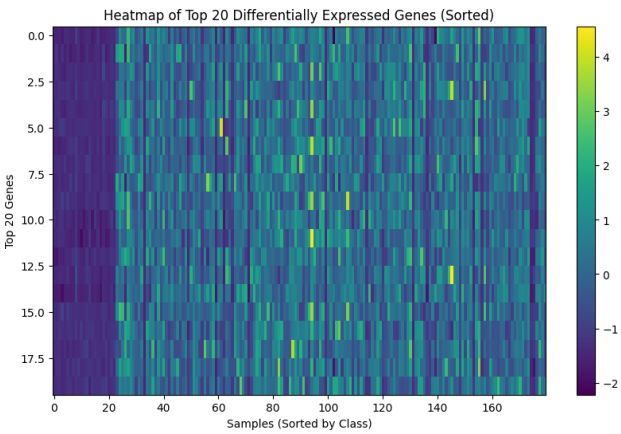


Figure 4. Heatmap of the top 20 differentially expressed genes: Identified through stability selection. Expression values were standardized across samples and sorted by class. Distinct clustering patterns demonstrate coordinated transcriptional divergence between glioma and non-tumor tissues.

7. Discussion

7.1 Summary of Findings

This study demonstrates that glioma tissue exhibits widespread transcriptional dysregulation relative to non-tumor brain tissue. Differential expression analysis identified 29,798 probes (approximately 54.6% of measured probes) as significantly dysregulated at FDR-adjusted $\alpha = 0.05$.

The presence of multiple genes with |Cohen's d| > 2 supports the magnitude of transcriptional divergence between tumor and control samples.

High classification performance under strict stratified cross-validation indicates robust separability.

7.2 Geometric Interpretation of Linear Separability

In high-dimensional feature spaces, data points become increasingly separable due to geometric expansion of representational space. As dimensionality increases, the probability of linear separability increases even when structure is subtle.

Linear SVM maximizes margin:

$$\text{Margin} = \frac{2}{\| w \|}$$

Maximizing margin improves generalization and reduces overfitting risk.

The superior performance of Linear SVM observed in this study aligns with transcriptomic classification literature (Statnikov et al., 2008).

In the present dataset ($p \approx 54{,}613$; $n = 180$), this geometric property likely contributes to the observed high performance of linear classifiers.

7.3 Biological Interpretation of Stable Genes

SOX2

A transcription factor regulating neural stem cell maintenance. SOX2 overexpression is associated with glioblastoma stem-like populations (Gangemi et al., 2009).

MSI2

An RNA-binding protein implicated in proliferation and oncogenic signaling (Kharas et al., 2010).

CDK17

Associated with cell cycle regulation and proliferative control (Sherr, 1996).

RAB Family Genes

Involved in vesicle trafficking and cellular signaling (Stenmark, 2009).

DNM3

Associated with membrane remodeling and invasive properties.

The convergence of statistical stability and biological plausibility strengthens confidence in the biological relevance of the identified markers.

7.4 Dimensionality Reduction Insights

Variance filtering achieved the highest mean accuracy (97.8% ± 3.24%), marginally exceeding PCA-based representation.

PCA reduced multicollinearity but transformed features into linear combinations of genes, reducing direct interpretability.

7.5 Model Stability and Generalization

Performance variance across folds was low for linear models, indicating stable generalization behavior.

Embedding preprocessing inside cross-validation ensured leakage-free evaluation.

Despite strong performance metrics, the absence of external validation limits conclusions regarding generalizability beyond the analyzed cohort.

7.6 Integrated Interpretation

Collectively, the results demonstrate that glioma and control brain tissues occupy distinct regions in high-dimensional gene expression space. Differential expression, dimensionality analysis, supervised classification, and stability selection converge to indicate structured and reproducible transcriptional divergence. The alignment between statistical robustness and biological coherence suggests that the observed classification performance reflects underlying molecular organization rather than modeling artifacts.

8. Limitations

Although classification performance was high, the absence of independent external validation limits conclusions regarding generalizability beyond the analyzed cohort. Validation using independent datasets or prospective samples would be necessary to confirm robustness across populations.

The dataset exhibits class imbalance (157 glioma vs 23 control samples), which may influence classifier behavior despite stratified cross-validation and class weighting adjustments. While performance metrics remained strong, imbalance can affect minority class sensitivity estimates.

This study relies exclusively on a single microarray platform (Affymetrix GPL570). Microarray technology has inherent limitations in dynamic range and probe specificity compared to RNA-sequencing approaches, potentially affecting detection sensitivity and transcript coverage.

Additionally, the analysis is correlational in nature. Differential expression and predictive modeling identify statistical associations but do not establish causal or mechanistic relationships between genes and glioma pathogenesis.

Finally, glioma subtypes (glioblastoma, oligodendroglioma, astrocytoma) were aggregated into a single tumor class. Although this aligns with the binary classification objective, subtype-specific molecular heterogeneity was not explicitly modeled and warrants further investigation.

9. Conclusion

This study provides a statistically rigorous, reproducible, and interpretable framework for glioma classification using high-dimensional gene expression data. After systematic metadata reconciliation and leakage-free validation, results demonstrate that transcriptomic data contain structured and reproducible disease-associated patterns capable of distinguishing glioma from non-tumor brain tissue with high accuracy.

Differential expression analysis identified 29,798 significantly dysregulated probes, with adjusted p-values reaching $< 1 \times 10^{-45}$ and multiple genes exhibiting large effect sizes ($|\text{Cohen's } d| > 2$). Among evaluated models, linear Support Vector Machines achieved the highest mean cross-validated accuracy (98.9% ± 1.36%) and ROC-AUC (0.996 ± 0.008), indicating strong and stable discriminative performance under stratified validation.

The integration of differential testing, dimensionality control, supervised learning, and stability selection provides a defensible methodological framework for analyzing high- dimensional transcriptomic datasets. The convergence of statistical robustness and biological plausibility suggests that the observed classification performance reflects underlying molecular organization rather than modeling artifacts.

Collectively, these findings support the hypothesis that human gene expression data contain stable, interpretable, and predictive signatures capable of distinguishing glioma from non-tumor brain tissue.

10.Future Work

Future research should prioritize independent external validation using additional glioma transcriptomic datasets to evaluate the generalizability of the proposed framework across populations and platforms.

Incorporating RNA-sequencing data may improve transcript coverage and dynamic range compared to microarray-based measurements. Comparative analysis across microarray and RNA-seq platforms would strengthen translational relevance.

Further investigation into subtype-specific classification (glioblastoma, oligodendroglioma, astrocytoma) may reveal additional molecular heterogeneity masked by the binary aggregation used in this study.

Pathway-level and gene set enrichment analyses could provide deeper biological insight into coordinated regulatory mechanisms underlying the identified core gene signature.

Finally, integrating multi-omics data (e.g., methylation, copy number variation, or proteomics) may enhance predictive modeling and contribute toward more comprehensive molecular characterization of glioma.

11. Data Availability Statement

Gene expression data are publicly available from the NCBI Gene Expression Omnibus (GEO) under accession number GSE4290 (Edgar et al., 2002).

12. Software & Reproducibility Statement

Analyses were conducted in Python (version 3.13.5) using scikit-learn (Pedregosa et al., 2011), NumPy, Pandas, Statsmodels, and Matplotlib. All preprocessing steps—including imputation, scaling, and dimensionality reduction—were embedded within stratified cross-validation pipelines to prevent data leakage and ensure reproducibility.

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14. Appendix

Appendix A: 3D PCA Plot

Rationale:
2D PCA is sufficient.
3D adds redundancy

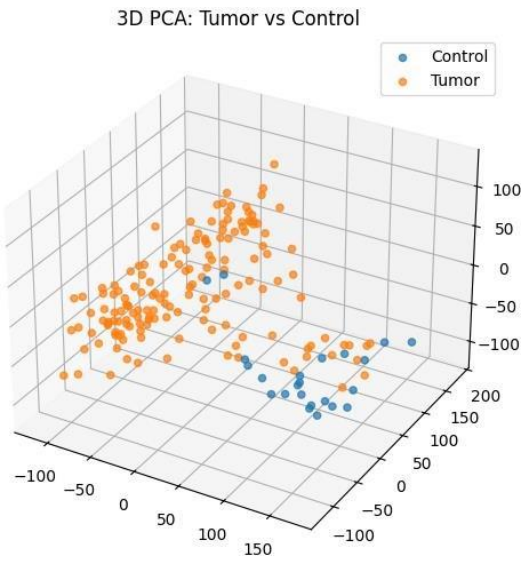

Figure A1. Three-dimensional PCA projection illustrating clustering consistency across the first three components.

Appendix B: Model Performance vs Feature Set Size.

Rationale:
It supports compression insight but not core claim.

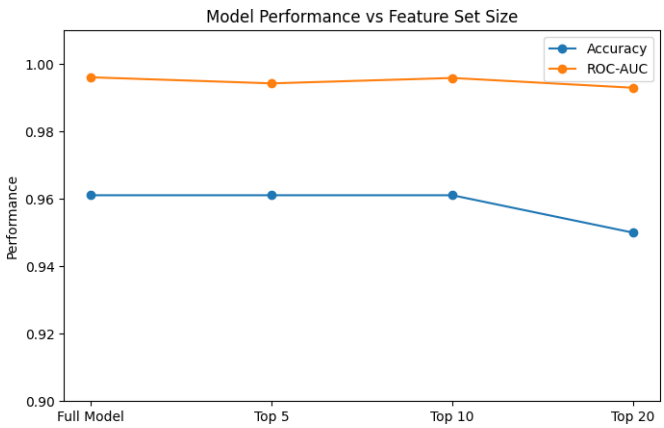

Figure B1. Classification accuracy and ROC-AUC across varying feature set sizes.